

06 · Is spend *causing* sales? — causal MMM (pymc-marketing)

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Runs in the legacy environment (pymc<6): `make env-legacy, kernel cmp-legacy.`

The decision. Brand search converts like crazy — pour budget into it? Careful: brand search mostly **catches** demand that upper-funnel channels (TV) and seasonality already created. **Last-click attribution over-credits it** and misallocates budget. A **causal MMM** encodes the marketing DAG so it separates *incremental* contribution from *mediated* credit — controlling for genuine confounders (seasonality) — then reports each channel’s contribution, **saturation curve**, and **ROI** with uncertainty, which we turn into a **budget reallocation**.

Two honest themes run through this notebook: (1) the confounding correction is robust and decisive; (2) MMM’s *absolute* levels are not — so you **calibrate them with a geo experiment** (Anchor B).

7-step contract.

1.1 2 · Simulate a ground truth

Weekly sales from a faithful **adstock** + **Michaelis-Menten saturation** process. **TV** is a big genuine driver. **seasonality** is a **confounder** — it lifts sales directly *and* drives brand-search spend. **brand_search** has only a *small* real effect, but its spend rides the seasonal wave, so a naive attribution ignoring seasonality massively over-credits it. Honest answer: TV highly incremental, brand_search only slightly.

TRUE incremental sales - TV 3,536 brand_search 333 (small)

	date_week	tv	brand_search	seasonality	sales
0	2023-01-02	42.870187	21.749830	0.000000	130.875186
1	2023-01-09	53.049712	18.348364	0.120537	136.412935
2	2023-01-16	27.623875	25.155597	0.239316	135.190674
3	2023-01-23	56.329747	32.172097	0.354605	152.617219
4	2023-01-30	43.339248	35.793793	0.464723	156.933276

1.2 3 · Identify — the marketing DAG picks controls vs mediators

DAG: `seasonality -> {sales, brand_search}, tv -> sales, brand_search -> sales`. The backdoor criterion says **seasonality is a confounder -> control for it**; omitting it is exactly the last-click mistake that over-credits brand search. The estimand for channel *c* is the interventional

contribution $\mathbb{E}[\text{sales} \mid \text{do}(x_c)] - \text{baseline}$; ROI = contribution $\div$ spend, each with a posterior. We hand the DAG to the MMM via `dag`, `treatment_nodes`, `outcome_node`.

Backdoor adjustment set the causal layer will use: `['seasonality']`

1.3 4 · Estimate — fit the causal MMM

A caution up front, and an honest one: **MMMs are notoriously miscalibrated in *absolute* terms** — with channel spend correlated with seasonal demand, a short weekly series can't fully separate “channel drove sales” from “season drove both”, so fitted contributions run *higher* than truth. That's why practitioners **calibrate MMMs with geo experiments** (nb 07). What the MMM *does* get right, and what drives budget, is the **ranking** and each channel's **ROI verdict**.

Output()

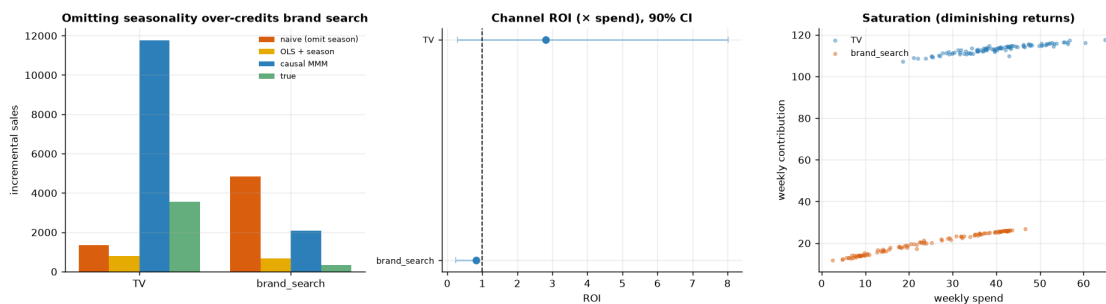
Causal MMM contribution - TV 11,775 brand_search 2,083
(true: TV 3,536, brand_search 333 - absolute levels run high; see note.)

1.4 5 · Validate — confounding correction, ROI with uncertainty, and saturation

Three views:

1. **The confounding correction** (robust): naive attribution *omitting* seasonality over-credits brand_search; adjusting for it (what the DAG prescribes) collapses that credit toward the truth.
2. **ROI with uncertainty** — a forest of each channel's posterior ROI vs the break-even line; this is the decision-relevant quantity, and it's far more stable than the absolute contribution.
3. **Saturation curves** — contribution vs spend per channel, showing diminishing returns (the shape that determines *marginal* ROI and hence where the next euro should go).

brand_search credit - naive 4,843 | adjusted 679 | true 333
ROI - TV 2.81x [90% 0.29,8.02] · brand_search 0.82x [90% 0.24,0.89]



1.5 6 · Decide, in euros — reallocate budget on marginal ROI

Budget chases **incremental ROI**, not last-click credit. Even with approximate absolute levels, the **ROI ordering is decisive**: brand_search sits below break-even (mostly capturing seasonal

demand), TV clears it. We estimate the sales impact of shifting a slice of budget from the low-ROI to the high-ROI channel — the concrete action — and flag that the *levels* need a geo-test calibration.

Shifting 15% of brand_search spend (€383) to TV -> projected +€764 sales

$P(\text{TV ROI} > \text{brand_search ROI}) = 0.96$ -> reallocate toward TV

Caveat: this uses AVERAGE ROI; with saturation, marginal ROI falls as you pour more into TV -

so reallocate in steps and re-estimate, and calibrate the absolute levels with a geo experiment.

1.6 7 · Caveats

- **Absolute contributions need calibration.** Here they ran ~2-3x the truth because channel spend rides the seasonal wave the MMM can't fully net out from a short series. **Trust the ranking and the ROI verdict, not the absolute euros** — calibrate levels with a geo experiment (nb 07). This is the single most important thing to say about any MMM.
- **The DAG is the assumption.** If TV genuinely creates incremental search value, encode that edge or you'll under-credit it.
- **Adstock/saturation priors matter.** Carry-over (λ) and diminishing-returns (α , κ) are weakly identified from limited weekly data; use informative priors from experiments where you have them.
- **Correlated channels -> wide, correlated ROI posteriors.** When two channels move together, the data can't fully separate them; the intervals will (honestly) show it.